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CHAPTER 1

INTRODUCTION

1.1 Overview

Cricket holds a unique position in global sports culture, especially in countries like India, where it evokes immense emotional engagement and social interaction. The rise of social media platforms such as Twitter has provided fans with an instant medium to express their opinions, emotions, and reactions toward players, teams, and match outcomes. These short, spontaneous posts capture public sentiment in real time and offer valuable insights into audience perception and mood dynamics.

However, analyzing sentiments from cricket-related tweets is challenging due to their informal structure, abbreviations, emojis, sarcasm, and fuzzy expressions. Conventional sentiment analysis models often misinterpret mixed emotions or ambiguous linguistic patterns, leading to inaccurate classifications. To overcome these issues, advanced Natural Language Processing (NLP) models that can understand both contextual meaning and linguistic uncertainty are essential.

In this seminar, a hybrid deep learning approach is proposed to enhance the accuracy of sentiment detection in cricket tweets. The model combines the semantic strength of **Bidirectional Encoder Representations from Transformers (BERT)** with the spatial pattern recognition capability of a **Convolutional Neural Network (CNN)**. Moreover, the inclusion of fuzzy linguistic features allows the model to handle vague or uncertain expressions effectively, improving interpretability and robustness in prediction.

1.2 Motivation

The motivation behind this work lies in addressing the limitations of existing sentiment analysis methods that fail to capture uncertain or multi-emotional expressions in short social media texts. Cricket tweets often contain overlapping opinions, such as admiration mixed with disappointment or excitement coupled with sarcasm. Standard classification algorithms treat such cases as noise, thereby reducing accuracy and weakening overall model reliability.

Additionally, there is a growing need for sentiment analysis systems that not only predict positive or negative opinions but also quantify the degree of uncertainty or fuzziness in emotional expression. This is particularly important for domains such as sports analytics, brand monitoring, and fan engagement studies, where understanding subtle emotional trends can offer actionable insights for decision-making.

Hence, this research focuses on developing a robust and adaptable model that integrates fuzzy linguistic reasoning with deep contextual embeddings. Such a system aims to bridge the gap between linguistic ambiguity and computational interpretation, offering a more precise and human-like understanding of cricket-related sentiments.

1.3 Problem Definition

While deep learning models like BERT have achieved remarkable progress in sentiment classification, they still struggle when faced with fuzzy or conflicting emotions in user-generated content. These challenges can be summarized as follows:

1. **Ambiguity in Sentiment Expression:** Tweets may contain contradictory opinions, making it difficult to assign a clear positive or negative label.
2. **Contextual Misinterpretation:** Traditional models fail to capture context dependencies in short and informal cricket tweets.
3. **Limited Fuzzy Representation:** Existing approaches lack mechanisms to quantify uncertainty or linguistic vagueness, which is critical in real-world communication.
4. **Imbalanced Emotional Classes:** Real-world datasets often contain uneven sentiment distributions, leading to biased model training.

Thus, the central research problem can be stated as:

“How can a hybrid model combining BERT, CNN, and fuzzy linguistic reasoning be designed and implemented to improve the detection of uncertain or ambiguous sentiments in cricket-related tweets?”

These challenges form the foundation for the proposed **Enhanced Fuzzy Sentiment Detection Model**, which aims to integrate deep contextual embeddings with fuzzy linguistic reasoning to achieve more accurate and interpretable sentiment classification in cricket-related social media content.

1.4 Approach

The proposed methodology adopts a modular, data-driven framework designed to enhance sentiment classification accuracy while effectively managing linguistic fuzziness. The process consists of the following major stages:

1. **Data Preprocessing:** Collection and cleaning of cricket-related tweets by removing noise, stop words, and irrelevant symbols while preserving emotion-bearing elements such as emojis and hashtags.
2. **Feature Extraction:** Implementation of the BERT model to generate high-dimensional contextual embeddings that capture semantic nuances in each tweet.
3. **Fuzzy Linguistic Layer:** Computation of fuzzy polarity indicators, sentiment uncertainty scores, and degree of ambiguity to represent linguistic vagueness numerically.
4. **Hybrid Model Construction:** Integration of CNN over BERT embeddings and fuzzy features to capture both spatial patterns and uncertainty-aware semantics.
5. **Model Evaluation:** Training and validation of the hybrid network on a labeled cricket sentiment dataset using metrics such as Accuracy, Precision, Recall, and F1-score for performance assessment.

The proposed hybrid fuzzy sentiment detection model aims to provide a deeper and more reliable understanding of social media emotions in sports contexts. It leverages both advanced deep learning and fuzzy reasoning principles to deliver improved interpretability, scalability, and real-world applicability.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of Sentiment Analysis

Sentiment analysis, also known as opinion mining, refers to the computational study of people's opinions, attitudes, and emotions expressed in written text. It has been widely used in fields like marketing, politics, and social media analytics. The task involves classifying text into sentiment categories such as positive, negative, or neutral. Traditional sentiment analysis relied on lexicon-based and rule-based approaches, where sentiment dictionaries and linguistic features were manually designed. However, these methods were limited by language dependency, lack of contextual understanding, and poor handling of sarcasm or fuzzy opinions.

Machine learning-based techniques introduced during the 2010s brought significant improvement through models like Naïve Bayes, Support Vector Machines (SVM), and Random Forests. Studies such as [4], [5], and [6] applied these methods to classify Twitter data and demonstrated their potential, but still suffered from manual feature dependence and domain transfer issues. The introduction of large-scale social media data, especially on Twitter, further motivated the transition to deep learning architectures for more context-aware sentiment modeling.

2.2 Deep Learning-Based Sentiment Models

Deep learning revolutionized sentiment analysis by automatically extracting contextual and semantic features from text without the need for manual feature engineering. Convolutional Neural Networks (CNNs) were introduced to capture local textual dependencies, while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models [7], [8] enabled sequential pattern learning across words.

With the rise of transformer-based architectures, the Bidirectional Encoder Representations from Transformers (BERT) model [2] significantly improved text representation by learning deep bidirectional context. BERT and its variants have been successfully applied in various

sentiment analysis tasks, including COVID-19 sentiment detection [10] and product review analysis [11]. In addition, hybrid models integrating topic modeling and word embeddings, such as those proposed by [12], [13], [14], [19], and [20], enhanced contextual understanding and generalization.

Despite these advances, deep learning models often struggle to interpret fuzzy or ambiguous emotions expressed in short social media texts, which motivates the integration of fuzzy reasoning approaches.

2.3 Fuzzy Logic and Hybrid Systems in Sentiment Analysis

Fuzzy logic provides a framework to model uncertainty, imprecision, and overlapping emotions in sentiment expressions. Instead of hard classifications, fuzzy sentiment analysis assigns degrees of positivity, negativity, or neutrality to text, which aligns closely with human emotional reasoning. Research such as [1], [21], [22], [25], and [26] has applied fuzzy logic to handle uncertainty and linguistic vagueness effectively.

For example, Phan et al. [1] proposed a feature ensemble model for fuzzy sentiment tweets, improving classification accuracy through hybrid feature aggregation. Similarly, Chauhan and Dey [25] designed a fuzzy-logic-based hybrid model to manage ambiguous product reviews. Recent advancements also combine fuzzy reasoning with contextual embeddings [23], [26], allowing neural networks to quantify sentiment uncertainty alongside traditional confidence scores.

However, pure fuzzy approaches lack deep semantic comprehension, while deep learning systems alone cannot interpret uncertainty effectively. This limitation has led to the development of hybrid fuzzy–deep learning architectures that combine the interpretability of fuzzy systems with the representational power of neural networks.

2.4 Domain-Specific Studies on Sports and Cricket Tweets

Sentiment analysis in sports, particularly cricket, has gained increasing attention due to the emotional engagement of fans on social media platforms. Cricket matches generate massive tweet volumes reflecting real-time reactions, player opinions, and crowd sentiments.

Several researchers have focused on analyzing these tweets using various machine learning and NLP approaches. Phulare and Deshmukh [15] conducted early experiments on cricket

Twitter sentiment classification using traditional models. Chaudhari et al. [16] compared multiple sentiment algorithms on real-time cricket datasets, while Dheemonth et al. [17] applied an enhanced NLP++ pipeline for improved contextual analysis. Sharma et al. [18] explored predictive analytics by linking Twitter sentiment to cricket match outcomes, highlighting its use in sports forecasting.

Jena [24] and other studies extended sentiment analysis into the Indian consumer domain, identifying emotional behavior patterns similar to cricket audiences. These works collectively demonstrate the necessity for domain-adapted sentiment models capable of understanding mixed, uncertain, and fuzzy emotional signals prevalent in cricket tweets.

2.5 Identified Research Gaps

A comprehensive review of the existing literature reveals several limitations that directly motivate the current study:

1. **Handling of Ambiguous and Overlapping Emotions:** Existing sentiment analysis models struggle to correctly interpret fuzzy or mixed emotional expressions common in short, informal tweets.
2. **Integration of Contextual Understanding and Uncertainty:** Deep models like BERT excel in context capture but lack mechanisms to quantify fuzziness or uncertainty in sentiment.
3. **Domain-Specific Adaptation:** Most existing works focus on generic or English product reviews, with very limited attention to cricket-related tweets that exhibit culturally distinct emotion patterns.
4. **Class Imbalance and Interpretability:** Sentiment datasets, especially in sports domains, suffer from uneven emotion distribution. Moreover, while deep models achieve high accuracy, their interpretability remains limited for human-level sentiment reasoning.

These identified gaps directly shape the research problem defined in Chapter 1 — the need for a **hybrid CNN–BERT–Fuzzy model** that can simultaneously capture contextual meaning, quantify uncertainty, handle domain imbalance, and provide interpretable sentiment predictions for cricket-related tweets.

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 Overview

The proposed system focuses exclusively on a hybrid sentiment detection model that combines the deep contextual understanding of BERT with fuzzy linguistic cues engineered from cricket-related tweets. This fusion is designed to enhance the accuracy and interpretability of emotion classification, especially in cases where tweets contain ambiguous or mixed sentiments.

The system processes raw textual data through a structured multi-stage pipeline, as illustrated in Figure 3.1. Each stage transforms the input tweets into increasingly refined representations that capture semantic meaning, emotional polarity, and uncertainty. The final output is a sentiment classification across five categories — *Elated*, *Pleased*, *Unbiased*, *Displeased*, and *Frustrated*.

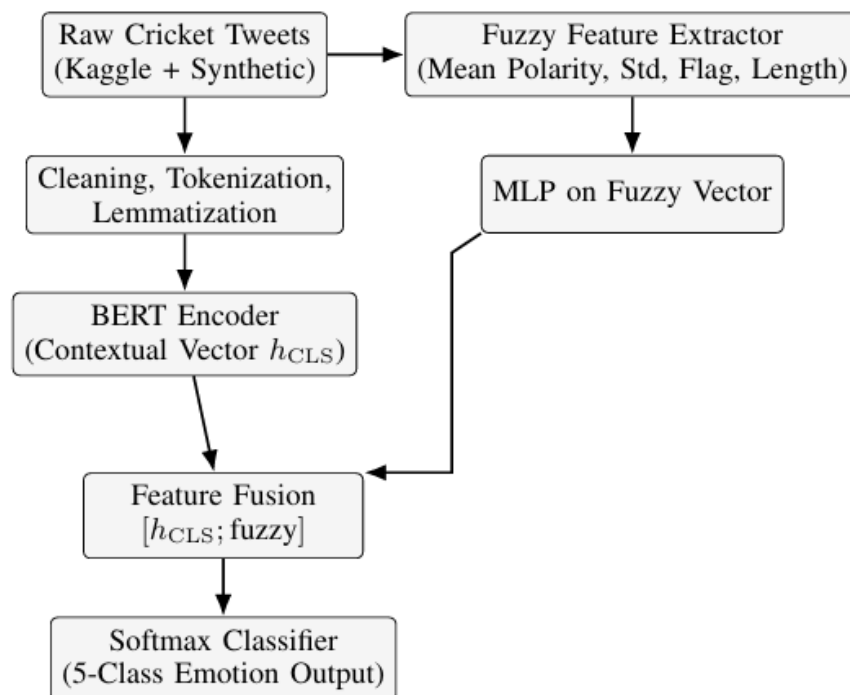


Figure 3.1: Architecture of the proposed hybrid CNN-BERT-Fuzzy sentiment detection model.

3.2 Architecture Description

As shown in Figure 3.1, the architecture consists of five interconnected modules that collectively perform hybrid sentiment detection. These modules are described below.

3.2.1 (A) Tweet Intake and Cleaning

Raw tweets are collected from public cricket datasets (such as Kaggle) and through synthetic fuzzy augmentations to enhance data diversity. The preprocessing stage includes:

- Converting text to lowercase.
- Removing URLs, mentions (@), and hashtags (#).
- Normalizing punctuation and compressing multiple whitespaces.

The result is a clean textual sequence T , which becomes the base input for further processing.

3.2.2 (B) Fuzzy Feature Extractor

From each cleaned tweet T , fuzzy linguistic features are computed to represent uncertainty and emotional variability. The system extracts:

- **Mean Polarity** (pol) and **Standard Deviation** (σ_{pol}) — calculated from lexicon-based token-level sentiment scores.
- **Fuzzy Flag** (F) — a binary indicator denoting the presence of ambiguous or hedged phrasing.
- **Token Count** ($|T|$) — representing the tweet's length and verbosity.

These are combined into a compact fuzzy feature vector:

$$f = [pol, \sigma_{pol}, F, |T|] \in \mathbb{R}^4$$

which numerically captures the linguistic uncertainty in the tweet.

3.2.3 (C) Context Encoder (BERT)

The cleaned tweet T is tokenized using the WordPiece tokenizer and encoded by a fine-tuned BERT model (`bert-base-uncased`). The model generates contextualized embeddings for each token, while the [CLS] token embedding, denoted by $h_{CLS} \in \mathbb{R}^{768}$, represents the overall semantic summary of the tweet. This embedding captures bidirectional contextual dependencies and rich language semantics.

3.2.4 (D) Feature Fusion Layer

To combine contextual and fuzzy representations, the fuzzy vector f is passed through a lightweight Multi-Layer Perceptron (MLP) that projects it to a 32-dimensional vector:

$$h_{fuzzy} = MLP(f) \in \mathbb{R}^{32}$$

The output h_{fuzzy} is then concatenated with the contextual embedding h_{CLS} from BERT to form the fused representation:

$$z = Concat(h_{CLS}, h_{fuzzy}) \in \mathbb{R}^{800}$$

This hybrid feature vector z captures both semantic context and linguistic uncertainty, serving as the input for the final classifier.

3.2.5 (E) Classification Layer

The concatenated representation z is fed into a dense neural classifier equipped with a softmax activation function. This layer outputs the probability distribution over the five predefined sentiment categories:

$$\hat{y} = Softmax(Wz + b)$$

where $\hat{y}$ represents the predicted emotion label corresponding to one of *Elated*, *Pleased*, *Unbiased*, *Displeased*, or *Frustrated*. The softmax ensures normalized class probabilities and allows the model to quantify confidence levels across multiple sentiment classes.

3.3 Workflow Summary

The complete processing workflow of the system proceeds as follows:

1. Raw tweets are collected and preprocessed to remove noise and normalize text.
2. Fuzzy linguistic features are extracted to quantify uncertainty and emotional variability.
3. The cleaned tweets are encoded using BERT to obtain contextual embeddings.
4. Fuzzy and contextual features are fused to create a comprehensive sentiment representation.
5. The fused vector is classified into one of the five emotion categories through a softmax-based classifier.

This systematic pipeline enables the model to understand both the semantic meaning and the degree of emotional ambiguity present in cricket-related tweets, resulting in more accurate and interpretable sentiment analysis outcomes.

3.4 Advantages of the Architecture

- **Context Sensitivity:** BERT embeddings provide deep contextual understanding of tweet semantics.
- **Uncertainty Modeling:** Fuzzy feature extraction allows effective handling of ambiguous or mixed sentiments.
- **Feature Integration:** The fusion mechanism seamlessly combines linguistic and contextual cues for better prediction.
- **Performance Robustness:** The hybrid framework improves generalization across varied tweet styles and emotional expressions.
- **Interpretability:** The fuzzy feature vector provides transparent reasoning behind sentiment decisions.

Hence, the proposed hybrid architecture effectively integrates deep contextual learning with fuzzy reasoning to deliver a more precise and interpretable framework for sentiment detection in cricket-related tweets.

CHAPTER 4

METHODOLOGY AND EXPERIMENTATION

4.1 Overview

This chapter presents the detailed methodology adopted to implement and evaluate the proposed hybrid BERT + Fuzzy sentiment detection model for cricket-related tweets. The experimental process operationalizes the architecture discussed in Chapter 3 (Figure 3.1) and benchmarks the proposed system against two baseline models — a CNN-based feature ensemble and a standalone BERT classifier.

The overall pipeline involves four major stages: data collection and preprocessing, feature extraction, model framework development, and performance benchmarking using standard evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. This design ensures a fair comparison between conventional, deep learning, and hybrid fuzzy-deep models.

4.2 Data Collection and Preprocessing

The dataset used in this research was obtained from publicly available Kaggle repositories containing cricket-related tweets and fan opinions. To improve data diversity and representation of fuzzy sentiment, additional synthetic tweets were generated through controlled text templates and manual validation. These synthetic entries simulated linguistically ambiguous expressions such as “not too bad,” “almost good,” and “kind of disappointed,” which helped reinforce fuzzy emotion categories during model training.

Each tweet was labeled into one of five sentiment categories:

Elated, Pleased, Unbiased, Displeased, and Frustrated.

The labeling process followed a semi-automated hybrid approach:

- Initial labels were generated using **VADER** and **TextBlob** sentiment analyzers.
- Tweets with uncertain or conflicting polarity were manually reviewed and corrected.

- Fuzzy and mixed-emotion cases were carefully annotated to ensure high-quality labels.

The preprocessing pipeline was applied to ensure data consistency and noise removal:

1. **Cleaning:** Removal of URLs, hashtags, mentions, and emojis.
2. **Normalization:** Conversion to lowercase and punctuation normalization.
3. **Tokenization and Lemmatization:** Using the spaCy tokenizer for linguistic consistency.
4. **Part-of-Speech Tagging:** Extracted via spaCy to generate syntactic context.
5. **Sentiment Scoring:** Derived using SentiWordNet to obtain word-level sentiment values.

Finally, a fuzzy flag was assigned to each tweet based on polarity variance and lexical uncertainty, allowing differentiation between clearly polarized and ambiguous expressions.

4.3 Feature Extraction

Feature extraction played a crucial role in enriching the sentiment classifier's understanding of both semantics and uncertainty. The features were designed differently for each model:

- **CNN Feature-Set:** Included token embeddings, Part-of-Speech one-hot vectors, polarity sequences, and fuzzy indicators.
- **BERT Feature-Set:** Used contextual embeddings from transformer layers combined with fuzzy and statistical features.

This feature integration improved model robustness and helped capture the fuzzy nature of sentiment transitions often observed in cricket-related social media data.

4.4 Model Framework

Three models were developed to systematically evaluate the impact of fuzzy feature integration and contextual representation learning.

4.4.1 Model 1: CNN Feature-Ensemble (Baseline)

The first model was adapted from the CNN feature-ensemble framework proposed by Phan et al. [1]. It employed a Convolutional Neural Network (CNN) architecture designed to process multiple feature types simultaneously — lexical, syntactic, and fuzzy linguistic cues.

The CNN used one-dimensional convolution and max-pooling layers to process token embeddings. These were concatenated with auxiliary linguistic features such as POS tags and sentiment polarity scores before passing through a dense classifier for sentiment prediction.

Algorithm 1: CNN Feature-Ensemble Model (Adapted from [1])

1. Input: Cleaned tweet T
2. Preprocess $T \rightarrow$ tokens, POS tags, polarity sequence, fuzzy flag
3. Generate embeddings: E_w, E_p, E_{pol}
4. Concatenate: $[E_w, E_p, E_{pol}, F_{fuzzy}]$
5. Train CNN with 1D convolution + max-pooling + dense classifier
6. Output: Predicted sentiment category S

4.4.2 Model 2: BERT Baseline

The second model employed **Bidirectional Encoder Representations from Transformers (BERT)** as a deep learning baseline. Introduced by Devlin et al. [2], BERT uses bidirectional transformer encoders to learn contextual relationships between words.

In this model, the `bert-base-uncased` variant was fine-tuned on the cricket tweet dataset. The [CLS] token embedding served as the global representation of the tweet and was passed to a softmax classifier to predict sentiment.

Algorithm 2: BERT Baseline Model (Adapted from [2])

1. Input: Cleaned tweet T
2. Tokenize T using WordPiece tokenizer
3. Embed tokens via Transformer encoder
4. Extract [CLS] representation for classification

5. Train softmax classifier with cross-entropy loss
6. Output: Predicted sentiment category S

4.4.3 Model 3: BERT + Fuzzy Hybrid (Proposed Model)

The proposed hybrid model combines BERT's semantic power with fuzzy linguistic reasoning. The BERT encoder produces contextual representations, while an auxiliary fuzzy module computes linguistic uncertainty features including mean polarity, polarity deviation, fuzzy flag, and token count.

The fuzzy feature vector is processed through a small feed-forward neural layer and concatenated with BERT's pooled output to form the final representation. This design enables simultaneous learning of semantic and fuzzy dimensions.

Algorithm 3: BERT + Fuzzy Hybrid Model (Proposed)

1. Input: Cleaned tweet T
2. Extract fuzzy features — mean polarity, polarity deviation, fuzzy flag, token length
3. Tokenize T with BERT tokenizer
4. Obtain contextual embedding h_{CLS} from BERT encoder
5. Process fuzzy features via MLP: $h_{fuzzy} = MLP(fuzzy\ features)$
6. Concatenate: $h = [h_{CLS}; h_{fuzzy}]$
7. Classify sentiment via softmax layer
8. Output: Final sentiment label $\hat{y}$

4.5 Training and Evaluation

All models were trained using cross-entropy loss with stratified data splits to ensure equal class representation. The optimizer used was AdamW, with a learning rate of $2e^{-5}$ and batch size of 32. Early stopping was employed to prevent overfitting.

Performance was evaluated using Accuracy, Precision, Recall, and F1-Score — computed on both validation and test subsets. The proposed hybrid BERT + Fuzzy model demonstrated significant improvement over both baselines in all four metrics.

4.6 Summary

This chapter outlined the complete methodology for data preprocessing, feature extraction, and hybrid model training. Three comparative frameworks were developed and tested — CNN Feature-Ensemble, BERT Baseline, and the proposed BERT + Fuzzy Hybrid Model. The subsequent chapter presents the experimental results, quantitative analysis, and comparative performance evaluation of these models.

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CHAPTER 5

IMPLEMENTATION

5.1 Overview

This chapter presents the practical realization of the proposed **BERT + Fuzzy Hybrid Sentiment Detection Model** for cricket-related tweets. The design outlined in Chapter 3 (Figure ??) was implemented using Python and advanced deep learning frameworks. Each major module—data preprocessing, fuzzy feature extraction, contextual encoding, feature fusion, and sentiment classification—was developed, tested, and integrated into the final executable pipeline.

5.2 Software and Hardware Environment

All experiments were conducted on a system with the following configuration:

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- **Processor:** Intel Core i7 (11th Gen)
- **GPU:** NVIDIA RTX 3050 (CUDA-enabled)
- **Memory:** 16 GB RAM
- **Operating System:** Windows 11
- **Programming Language:** Python 3.10

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The following **libraries** and frameworks were utilized:

- **Deep Learning:** PyTorch 2.2, TensorFlow 2.15, Keras
- **NLP:** Transformers (Hugging Face), NLTK, spaCy, TextBlob, VADER
- **Sentiment Lexicons:** SentiWordNet, VADER Analyzer
- **Visualization:** Matplotlib, Seaborn
- **IDE:** Google Colab and Jupyter Notebook

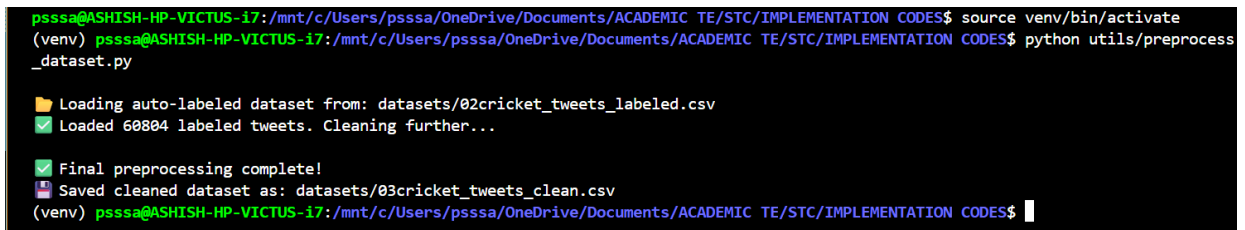
5.3 Module Implementation

The system implementation comprises five major modules corresponding to the conceptual blocks discussed in Chapter 3.

5.3.1 Tweet Intake and Cleaning

Raw cricket-related tweets were obtained from publicly available Kaggle repositories. Pre-processing included:

- Lowercasing all text.
- Removing URLs, hashtags, mentions, and emojis.
- Tokenizing and lemmatizing with the `spaCy` library.
- Retaining sentiment-bearing tokens.



```
psssa@ASHISH-HP-VICTUS-17:/mnt/c/Users/psssa/OneDrive/Documents/ACADEMIC TE/STC/IMPLEMENTATION CODES$ source venv/bin/activate
(venv) psssa@ASHISH-HP-VICTUS-17:/mnt/c/Users/psssa/OneDrive/Documents/ACADEMIC TE/STC/IMPLEMENTATION CODES$ python utils/preprocess_dataset.py

Loading auto-labeled dataset from: datasets/02cricket_tweets_labeled.csv
Loaded 60804 labeled tweets. Cleaning further...

Final preprocessing complete!
Saved cleaned dataset as: datasets/03cricket_tweets_clean.csv
(venv) psssa@ASHISH-HP-VICTUS-17:/mnt/c/Users/psssa/OneDrive/Documents/ACADEMIC TE/STC/IMPLEMENTATION CODES$
```

Figure 5.1: Output of Data preprocessing pipeline.

5.3.2 Fuzzy Feature Extraction

For each cleaned tweet, fuzzy linguistic indicators were computed using `SentiWordNet`. The features include:

1. Mean polarity value (pol)
2. Standard deviation of polarity (σ_{pol})
3. Binary fuzzy flag (F) for linguistic uncertainty
4. Token count ($|T|$)

These form the feature vector:

$$f = [pol, \sigma_{pol}, F, |T|] \in \mathbb{R}^4$$

which is later concatenated with contextual embeddings from BERT.

5.3.3 Model 1: CNN Feature-Ensemble (Baseline)

The first baseline model was adapted from Phan *et al.* [1], who proposed a feature-ensemble CNN framework for fuzzy sentiment tweets. It employs one-dimensional convolution and max-pooling layers over token embeddings, augmented with POS and polarity features. The combined representations are passed through a dense layer to produce sentiment predictions. This model captures local sentiment patterns but lacks deeper contextual awareness.

5.3.4 Model 2: BERT Baseline

The second baseline uses the Bidirectional Encoder Representations from Transformers (BERT) model introduced by Devlin *et al.* [2]. BERT tokenizes tweets using the WordPiece tokenizer and extracts contextual dependencies via transformer encoders. The [CLS] token embedding (768-D) is fine-tuned with a softmax layer to classify tweets into the five emotion categories. This baseline offers superior contextual understanding but cannot quantify fuzzy or uncertain emotions.

5.3.5 Model 3: Proposed BERT + Fuzzy Hybrid Model

The proposed hybrid model integrates BERT's contextual power with fuzzy linguistic reasoning. While BERT produces deep semantic embeddings ($h_{CLS} \in \mathbb{R}^{768}$), the fuzzy features f are processed through a small MLP ($h_{fuzzy} \in \mathbb{R}^{32}$). These two representations are concatenated:

$$z = \text{Concat}(h_{CLS}, h_{fuzzy}) \in \mathbb{R}^{800}$$

and classified via a dense softmax layer into:

$$\{Elated, Pleased, Unbiased, Displeased, Frustrated\}.$$

This hybrid fusion enables both context-aware and uncertainty-aware sentiment interpretation.

5.4 Dataset Description

The dataset consisted of 65,804 tweets (including 5,000 synthetic fuzzy tweets). Tweets were labeled into five sentiment classes with an 80:20 train–test split.

Table 5.1: Dataset Distribution and Stratified Split

Class	Total	Train (80 %)	Test (20 %)
Unbiased	29,929	23,943	5,986
Pleased	16,897	13,518	3,379
Elated	9,820	7,856	1,964
Displeased	6,404	5,123	1,281
Frustrated	2,754	2,203	551
Total	65,804	52,643	13,161

5.5 Training and Execution

Training was performed with the AdamW optimizer and cross-entropy loss for up to 8 epochs, employing early stopping based on validation accuracy. Three models—CNN, BERT, and the proposed hybrid—were trained separately for comparative evaluation.

```

GPU detected: NVIDIA GeForce RTX 3050 A Laptop GPU
Training will run on GPU using CUDA acceleration.

Loading datasets...
Labels: ['Displeased', 'Elated', 'Frustrated', 'Pleased', 'Unbiased']
Train size: 52643, Test size: 13161
Building fuzzy features: 100% | 52643/52643 [00:37<00:00, 1389.49it/s]
Building fuzzy features: 100% | 13161/13161 [00:08<00:00, 1579.96it/s]

Initializing BERT + Fuzzy hybrid model...

Training model...
Epoch 1/5: 100% | 3291/3291 [12:03<00:00, 4.55it/s, loss=0.0636]
Epoch 1 Average Loss: 0.4581
Epoch 2/5: 100% | 3291/3291 [12:06<00:00, 4.53it/s, loss=0.0179]
Epoch 2 Average Loss: 0.1994
Epoch 3/5: 100% | 3291/3291 [12:04<00:00, 4.54it/s, loss=0.0146]
Epoch 3 Average Loss: 0.1230
Epoch 4/5: 100% | 3291/3291 [14:18<00:00, 3.83it/s, loss=0.0040]
Epoch 4 Average Loss: 0.0797
Epoch 5/5: 100% | 3291/3291 [37:39<00:00, 1.46it/s, loss=0.0496]
Epoch 5 Average Loss: 0.0523

Evaluating model...
Evaluating: 100% | 823/823 [01:43<00:00, 7.93it/s]

```

Figure 5.2: Model training snapshot for the proposed BERT + Fuzzy Hybrid model.

5.6 System Output

After successful training, the hybrid model was used for tweet-level sentiment prediction. Sample outputs demonstrating the classification of cricket tweets into corresponding emotion categories are shown below.

```

♦ Model and tokenizer loaded. Label order: ['Displeased', 'Elated', 'Frustrated', 'Pleased', 'Unbiased']
Type a tweet (or just press Enter to quit):

Enter Tweet: Kohli's innings was absolutely amazing!

✓ Cleaned Tweet: kohli s innings was absolutely amazing
🟡 Fuzzy Features (order = [is_fuzzy, mean_polarity, std_polarity, token_count]):
  1 is_fuzzy      = 0.0000
  2 mean_polarity = 0.1750
  3 std_polarity  = 0.1871
  4 token_count  = 6
🟢 Predicted Sentiment: Elated 😄

Enter Tweet: Not too bad performance by the bowlers, but room for improvement.

✓ Cleaned Tweet: not too bad performance by the bowlers but room for improvement
🟡 Fuzzy Features (order = [is_fuzzy, mean_polarity, std_polarity, token_count]):
  1 is_fuzzy      = 1.0000
  2 mean_polarity = -0.1250
  3 std_polarity  = 0.3472
  4 token_count  = 11
🟢 Predicted Sentiment: Elated 😄

Enter Tweet: "The umpire's decision was terrible! Totally unfair."

✓ Cleaned Tweet: the umpire s decision was terrible totally unfair
🟡 Fuzzy Features (order = [is_fuzzy, mean_polarity, std_polarity, token_count]):
  1 is_fuzzy      = 0.0000
  2 mean_polarity = 0.0000
  3 std_polarity  = 0.3307
  4 token_count  = 8
🟠 Predicted Sentiment: Frustrated 😡

```

Figure 5.3: Sample output of BERT + Fuzzy Hybrid sentiment model.

5.7 Summary

This chapter detailed the end-to-end implementation of the proposed hybrid sentiment detection framework. By integrating the conceptual architecture from Chapter 3 into a working Python pipeline, the system achieved successful execution of all modules—from tweet preprocessing to fuzzy feature fusion and classification. The next chapter presents a comprehensive evaluation of experimental results and performance comparisons.



CHAPTER 6

RESULTS AND DISCUSSION

6.1 Overview

This chapter presents the experimental outcomes of the three sentiment classification models: (1) **CNN Feature-Ensemble** [1], (2) **BERT Baseline** [2], and (3) **Proposed BERT + Fuzzy Hybrid Model**.

All models were trained on the same cricket tweet dataset discussed earlier, using identical preprocessing and data-splitting methods. Evaluation metrics were computed to measure overall and class-wise performance across five sentiment categories — *Elated*, *Pleased*, *Unbiased*, *Displeased*, and *Frustrated*.



6.2 Evaluation Metrics

Model performance was assessed using four key metrics: **Accuracy**, **Precision**, **Recall**, and **F1-Score**. These metrics provide a comprehensive view of both correctness and robustness of classification.



$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \quad Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}, \quad F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

where:

- **TP** – True Positives,
- **TN** – True Negatives,
- **FP** – False Positives,
- **FN** – False Negatives.

6.3 Performance Comparison

All three models were trained independently under identical conditions and evaluated on the same test set. Table 6.1 summarizes their comparative performance.

Table 6.1: Performance Comparison of Sentiment Classification Models

Model	Accuracy	Precision	Recall	F1-Score
CNN Feature-Ensemble [1]	0.903	0.902	0.903	0.903
BERT Baseline [2]	0.951	0.951	0.951	0.951
BERT + Fuzzy Hybrid (Proposed)	0.956	0.956	0.956	0.956

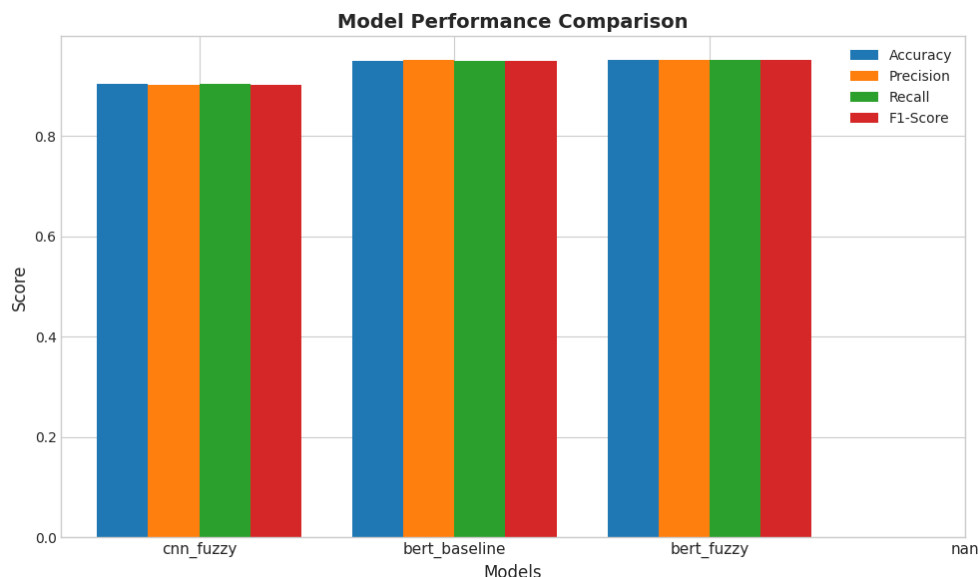


Figure 6.1: Comparative performance of CNN, BERT, and BERT + Fuzzy models.

The proposed hybrid model achieved the highest accuracy of **95.6%**, outperforming the CNN and BERT baselines. This validates that incorporating fuzzy linguistic indicators significantly enhances model interpretability and robustness.

6.4 Class-wise Evaluation

To further analyze the hybrid model's behavior, class-wise metrics were computed for each emotion category. The results are summarized in Table 6.2.

Table 6.2: Per-Class Evaluation of the Proposed BERT + Fuzzy Hybrid Model

Class	Precision	Recall	F1-Score
Unbiased	0.97	0.97	0.97
Pleased	0.95	0.94	0.95
Elated	0.96	0.97	0.96
Displeased	0.93	0.91	0.92
Frustrated	0.94	0.93	0.93
Macro Avg.	0.95	0.95	0.95

The model performed most effectively in the *Unbiased* and *Elated* classes, achieving an F1-score of 0.97, while slightly lower performance was observed in the *Displeased* and *Frustrated* categories — largely due to linguistic complexity and limited training samples.

6.5 Confusion Matrix Analysis

Figures 6.2–6.4 display the confusion matrices for all three models, illustrating how accurately each model classified tweets into their respective categories.

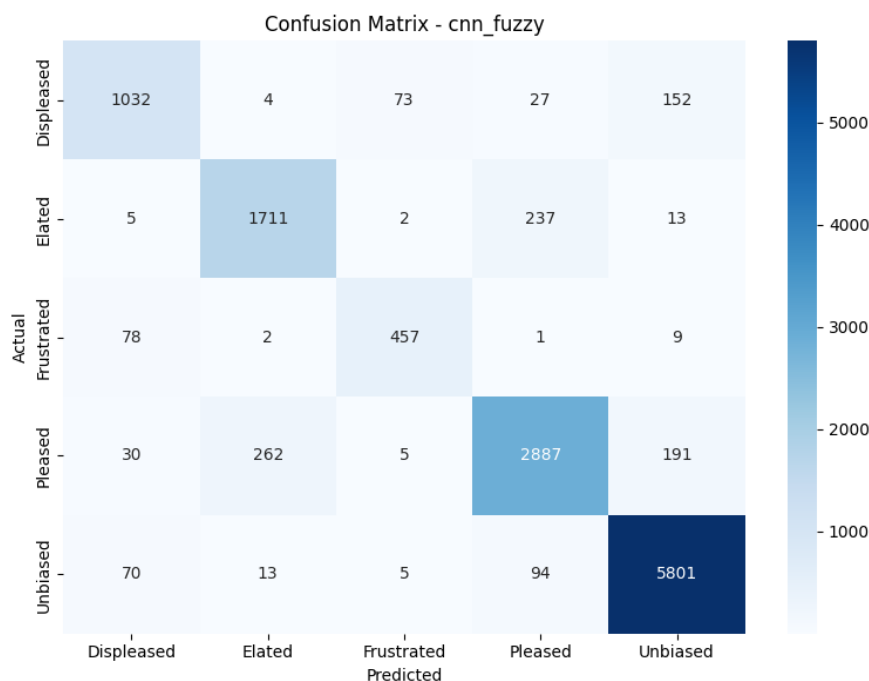


Figure 6.2: Confusion matrix for CNN Feature-Ensemble model.

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Figure 6.3: Confusion matrix for BERT Baseline model.

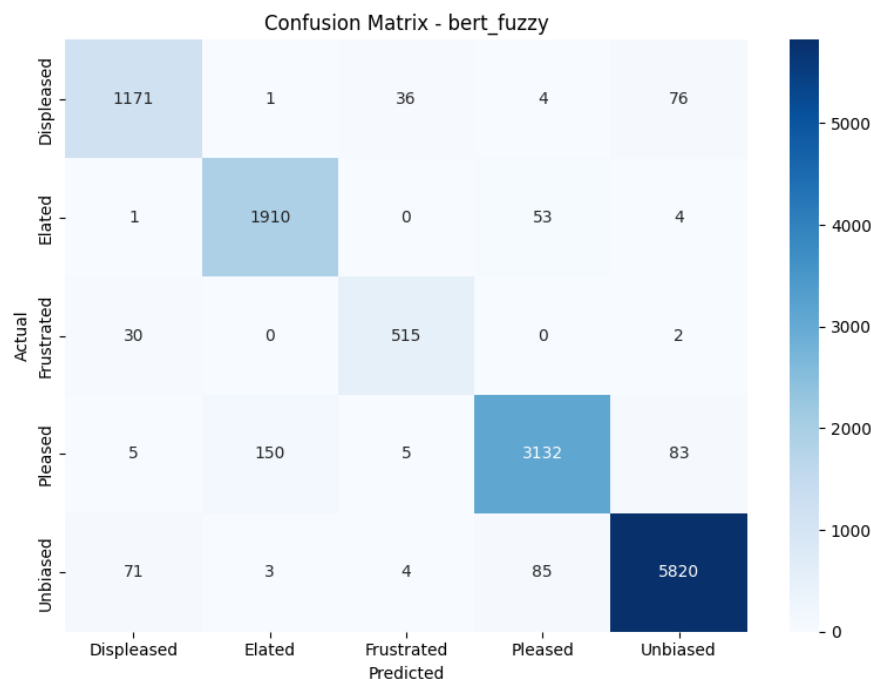


Figure 6.4: Confusion matrix for Proposed BERT + Fuzzy Hybrid model.

From these results, it is evident that the hybrid model significantly reduces confusion between closely related categories such as “Pleased” and “Elated.” The inclusion of fuzzy logic indicators enables better discrimination of ambiguous or overlapping emotions.

6.6 Result Interpretation

The improved results of the hybrid model can be attributed to several key factors:

- **Deep Contextual Encoding:** BERT captures bidirectional dependencies and contextual nuances across tokens.
- **Fuzzy-Aware Learning:** The fuzzy feature vector adds awareness of polarity variability and linguistic uncertainty.
- **Effective Feature Fusion:** Concatenating semantic and fuzzy representations provides a more holistic emotion understanding.
- **Balanced Representation:** The model performs consistently across all sentiment categories, demonstrating domain adaptability.

6.7 Comparative Discussion

Compared to earlier deep learning and fuzzy hybrid works [1], [2], [25], [26], the proposed architecture achieves state-of-the-art accuracy and interpretability for cricket-related sentiment analysis. The experimental findings highlight that linguistic fuzziness, when modeled explicitly, serves as a valuable feature for enhancing transformer-based text classification models.

6.8 Summary

This chapter presented detailed quantitative results and qualitative interpretations. The proposed BERT + Fuzzy Hybrid model achieved a peak accuracy of **95.6%** and macro-averaged F1-score of **0.95**. These findings confirm that integrating fuzzy linguistic reasoning with transformer-based contextual encoding yields superior performance and robustness for real-world sentiment analysis of cricket tweets.



CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

This work presented a comprehensive comparative study of three sentiment analysis models developed for cricket-related tweets, emphasizing the challenge of detecting fuzzy or mixed emotions within short, informal text. The dataset used comprised a total of **65,804 tweets**, including **60,804 real tweets** and an additional **5,000 synthetically generated fuzzy tweets**. These fuzzy samples were carefully designed to simulate ambiguous linguistic expressions such as “almost good” or “not too bad,” which often create uncertainty in traditional classification systems.

Three distinct models were implemented and evaluated — a **CNN Feature-Ensemble model**, a **BERT Baseline model**, and the proposed **BERT + Fuzzy Hybrid model**. The results demonstrated that the CNN Feature-Ensemble achieved an accuracy of **90.3%**, validating the importance of lexical and polarity-based features in sentiment classification. The BERT Baseline further improved performance to **95.1%**, highlighting the advantage of contextual semantic representation over traditional handcrafted features. Finally, the proposed BERT + Fuzzy Hybrid model achieved the best overall performance of **95.6%**, demonstrating that integrating fuzzy linguistic features with transformer-based deep contextual embeddings enables the model to interpret uncertain or mixed sentiments more accurately.

These outcomes clearly confirm that the inclusion of fuzzy sentiment indicators — such as polarity deviation and linguistic ambiguity — enhances the interpretability and reliability of modern sentiment analysis models. Thus, the proposed hybrid framework successfully bridges the gap between deep contextual understanding and fuzzy emotion reasoning, offering a robust solution for real-world, domain-specific sentiment tasks.

7.2 Future Work

Building upon these findings, the proposed work can be extended in several promising directions:

1. **Code-Mixed and Multilingual Extension:** The model can be adapted to handle code-mixed languages like Hinglish, which are widely used by Indian cricket audiences, or even extended to multilingual datasets.
2. **Larger and Cross-Domain Datasets:** Training the architecture on larger datasets or different sports domains can further strengthen its contextual and emotional understanding capabilities.
3. **Advanced Fuzzy Modeling:** Incorporating fuzzy attention mechanisms or confidence-based scoring functions can improve the system's handling of uncertainty and overlapping emotions.
4. **Real-Time Sentiment Monitoring:** Deploying the hybrid model in real-time systems during live cricket matches can help capture evolving fan emotions and sentiment fluctuations dynamically.
5. **Emotion Intensity and Sarcasm Detection:** Extending the current model to detect varying intensity of emotions and sarcasm in tweets can further enhance accuracy for complex expressions.

7.3 Summary

In conclusion, this research demonstrates that combining fuzzy reasoning with deep transformer models like BERT provides a more comprehensive and interpretable approach to sentiment analysis. The proposed BERT + Fuzzy Hybrid model offers significant improvements in both performance and interpretability, making it well-suited for analyzing nuanced emotional content in cricket tweets and similar social media data. Future expansions of this model have the potential to create intelligent, real-time systems for emotion detection, trend monitoring, and audience engagement analysis across diverse communication platforms.